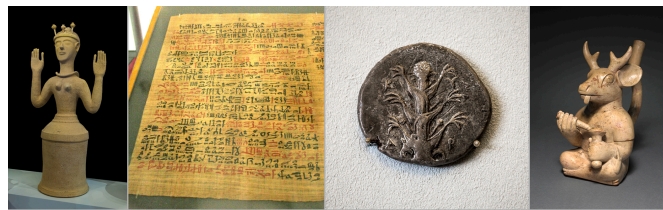


How do we Identify Drugs Archaeologically?



How do we Identify Drugs Archaeologically?

Many of these drugs have a long history documented in material evidence that has been found archaeologically. One of the most valuable is iconographic or written accounts of the use of particular drugs. Balick and Cox provide the example of opium-poppy capsule headdresses worn by female figures rendered in clay from Minoan Crete (modern Greece; c. 1400–1100 BCE). Egyptian and Sumerian textual records also cite the use of medicinal plants, including opium. Another medicinal plant, known to the ancient Greeks as silphium, became so popular as to be adopted as the image of the city of Cyrene, in what is today Libya, on its silver coins. South American societies depicted coca-leaf use in many figural representations, even those of human-animal hybrids.



From left to right: terracotta Minoan goddess figurine with poppy stalks on her headdress; an Egyptian medical text, the *Ebers Papyrus*, that includes mention of opium and cannabis as medical treatments; a silver tetradrachm coin of Cyrene, 435–330 BCE, depicting the silphium plant; Moche ceramic pot shaped as a deer-human hybrid person using coca, holding a lime flask and with a bag of coca leaves on the left forearm

Source: Wikimedia Commons—[Minoan Goddess](#)

(https://commons.wikimedia.org/wiki/File:Minoan_praying_women_archmus_Heraklion_Crete_Greece.jpg); [Ebers Papyrus](#)

(https://commons.wikimedia.org/wiki/File:PEbers_c41-bc.jpg); [Tetradrachm Coin](#)

(https://commons.wikimedia.org/wiki/File:Kyrene_-_435-330_BC_-_silver_tetradrachm_-

[_silphium_plant_-_head_of_Zeus_Ammon_-_Berlin_MK_BM_18200186.jpg](#); [Moche Ceramic Pot](#)
(https://commons.wikimedia.org/wiki/File:Clevelandart_2008.1.jpg).

Less obvious, but more widespread, are the remains of durable tools and containers used to store narcotics, or of narcotic plants themselves. These include tools such as the coca bag and flasks depicted above, but also cannabis-burning paraphernalia used by indigenous groups of the central Asian steppe, such as the Scythians. Burials of the Scythians and other central and west Asian societies include cannabis seeds that were placed within burials—in some cases, in large quantities. In 2016, a burial was published (<https://link.springer.com/article/10.1007/s12231-016-9351-1>) from the site of Jiayi, in what is today Xinjiang in western China, dated by radiocarbon to c. 800 BCE; in this burial, a human body was covered with 13 entire female cannabis plants, which were preserved exceptionally well due to the hyperarid setting of this cemetery.

The following contains images of human skeletal remains from an archaeological excavation. It is not necessary to view these images to complete this course.

The following contains images of human skeletal remains from an archaeological excavation. It is not necessary to view these images to complete this course. To view these images, please click the button below:

☒ **A. Correct:** Yes, I would like to view these images.

Correct answer

☐ **B.** No, I would **not** like to view these images.



A final source of archaeological evidence is chemical residues of narcotics found in containers or in human remains. Using the methods described for residue analysis in an earlier module, such as GC-MS, it is possible to identify chemicals produced by the metabolism of specific psychoactive drugs in the human body. This is much the same as how drug tests work today, although since we do not have access to blood in ancient bodies, such analyses are done on other soft tissues which are preserved under exceptional circumstances, such as the famous "Iceman" known as Ötzi (<https://en.wikipedia.org/wiki/%C3%96tzi>) found in a melting glacier in the Alps or more numerous natural mummies found when humans are buried in hyperarid regions, such as the coastal deserts of Peru. These methods are often used together to determine the presence of drugs in archaeological contexts.

